SLIIT UNI
THE KNOWLEDGE UNIVERSITY

IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

Project ID: **J26-IT-403**

1. Topic (12 words max)

Personalized Grooming Recommendation System with AI Assistance

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature and Gravity of the Problem

Many Sri Lankan young adults aged 18–35 face uncertainty when selecting hairstyles or grooming styles that suit their facial structure. They often rely on social media trends, peer opinions, or trial and error, yet these choices may not match their face shape, proportions, or personal features. Existing grooming applications and style guides are mostly designed around Western facial baselines and generic face-shape categories, which reduces their usefulness for South Asian users. As a result, users may not know their face shape, may choose unsuitable styles, and may make grooming decisions without reliable guidance. To address this gap, the proposed system provides AI-assisted facial analysis and personalized grooming recommendations tailored for South Asian, especially Sri Lankan, users.

- ***Primary users:*** *Sri Lankan young adults (18–35 years old) who want personalized hairstyle and grooming guidance before making grooming decisions.*
- ***Secondary users:*** *Barbershops, salons, and personal stylists who can use the system as a decision-support tool during customer consultations.*

This problem affects a broad group in society rather than a single individual or institution. Existing global tools are often not calibrated for South Asian facial characteristics, so they may produce less accurate or less relevant suggestions for Sri Lankan users.

Impact of the Problem

***Socially**, the system helps young adults understand their face shape and choose styles that better suit their appearance instead of blindly following trends. **Economically**, it supports salons, barbershops, and grooming businesses by improving consultation quality and customer satisfaction.*

FoC Research Cluster and Industry Verticals

*This project aligns with **CoEAI** by applying AI at multiple stages of facial analysis, feature scoring, and personalized recommendation generation. It also uses XAI to provide transparent reasoning behind each suggestion, allowing users to understand why a style is recommended.*

Alignment with Sustainable Development Goals (SDGs)

*The proposed research directly supports **SDG 3 (Good Health and Well-being)** by helping users make more confident and informed grooming decisions, which can contribute to personal well-being and self-image. It also contributes to **SDG 9 (Industry, Innovation and Infrastructure)** through the use of AI-based facial analysis, personalized recommendation logic, and a practical digital grooming solution that applies innovation to everyday needs. In addition, the project aligns with **SDG 10 (Reduced Inequalities)** by focusing on South Asian and Sri Lankan facial characteristics, addressing the gap created when global grooming tools are built mainly around Western or East Asian face types. This makes the system more inclusive and better suited to users who are often underrepresented in existing grooming technologies.*

References:

1. Raveendran, M., 2019. The South Asian facial anthropometric profile: A systematic review. Journal of Cranio-Maxillofacial Surgery, 47(2), pp.263-272.

Link: https://pubmed.ncbi.nlm.nih.gov/30573375/

2. Prasanna, L.C., et al. (2024). *Anthropometric Analysis of Indian Faces: A Systematic Review*.

Link: https://pubmed.ncbi.nlm.nih.gov/38911415/

3. Tio, A.E. (2019). *Face shape classification using Inception v3*.

Link: https://arxiv.org/abs/1911.07916

4. *A hybrid approach to building face shape classifier for hairstyle recommender system*.

Link: https://www.sciencedirect.com/science/article/pii/S0957417425027320

5. Marinescu, A.-I. (2021). *Automatic Face Shape Classification Via Facial Landmark Measurements*.

Link: https://discovery.researcher.life/article/automatic-face-shape-classification-via-facial-landmark-measurements/272051714de53eb2a

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Products and Systems

Existing commercial grooming systems include ***YouCam Makeup*** *by* ***Perfect Corp****,* ***L'Oréal Style My Hair*** *based on* ***ModiFace technology****, and* ***Beardify*** *by* ***Pinatsu Studio****. These applications provide attractive virtual try-on experiences by letting users preview hairstyles, hair colors, or beard styles on their own images. However, they are primarily designed for entertainment, marketing, or basic style experimentation rather than for structured personalized grooming guidance. Their recommendations are generally broad and visually driven, with limited support for deeper facial analysis or user-specific decision-making.*

Summary of Existing Research

Academic research has also made important progress in facial analysis and hairstyle recommendation. The paper ***"Integrated Multi-Model Face Shape Identification Framework"*** *presents a multi-model approach for face shape classification. The study* ***"A Hybrid Approach to Building Face Shape Classifier for Hairstyle Recommender System"*** *by combining* ***Multiple Kernel Learning (MKL)*** *and* ***VGG-Face*** *shows how deep learning can support hairstyle recommendation. More recently,* ***"Stable-Hair: Real-World Hair Transfer via Diffusion Model"*** *and* ***"HairPort: In-context 3D-aware Hair Import and Transfer for Images"*** *demonstrate advanced hair synthesis and transfer capabilities for realistic visual transformations. These works provide strong technical foundations, but many remain either isolated research models or limited proof-of-concept systems.*

Gaps and Limitations Identified

Our proposed system addresses these gaps by focusing on ***Sri Lankan and broader South Asian facial characteristics****, which are often underrepresented in existing apps and studies. Unlike commercial systems that mainly show visuals, our solution combines* ***facial profiling, localized analysis, personalized ranking, and explainable AI*** *to produce more accurate and understandable recommendations. Unlike research papers that often stop at classification or image generation, our system integrates these ideas into a complete, practical, user-oriented grooming recommendation platform. This makes the solution more personalized, more explainable, and more suitable for real-world use in the South Asian context.*

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Addressing the Identified Gaps

*The proposed **Personalized Grooming Recommendation System with AI Assistance** addresses the major gaps identified in existing commercial systems and academic studies by moving beyond generic style overlays and isolated face-shape classification models. Current applications mainly focus on visual experimentation, while many research works stop at classification, synthesis, or proof-of-concept stages. In contrast, our system is designed as a practical, user-oriented recommendation platform that focuses on South Asian facial characteristics, especially Sri Lankan users. It combines facial analysis, localized interpretation, style ranking, and explainable reasoning to provide personalized grooming guidance rather than only visual transformation. This makes the solution more accurate, more contextual, and more useful for real-world grooming decisions.*

System Perspective

Input:

The system begins with a user selfie or camera image. Before analysis, it checks whether the image is suitable for processing by assessing clarity, face visibility, and frontal alignment. Once the input is validated, the face region is prepared for further processing so that only reliable facial information is passed into the next stages.

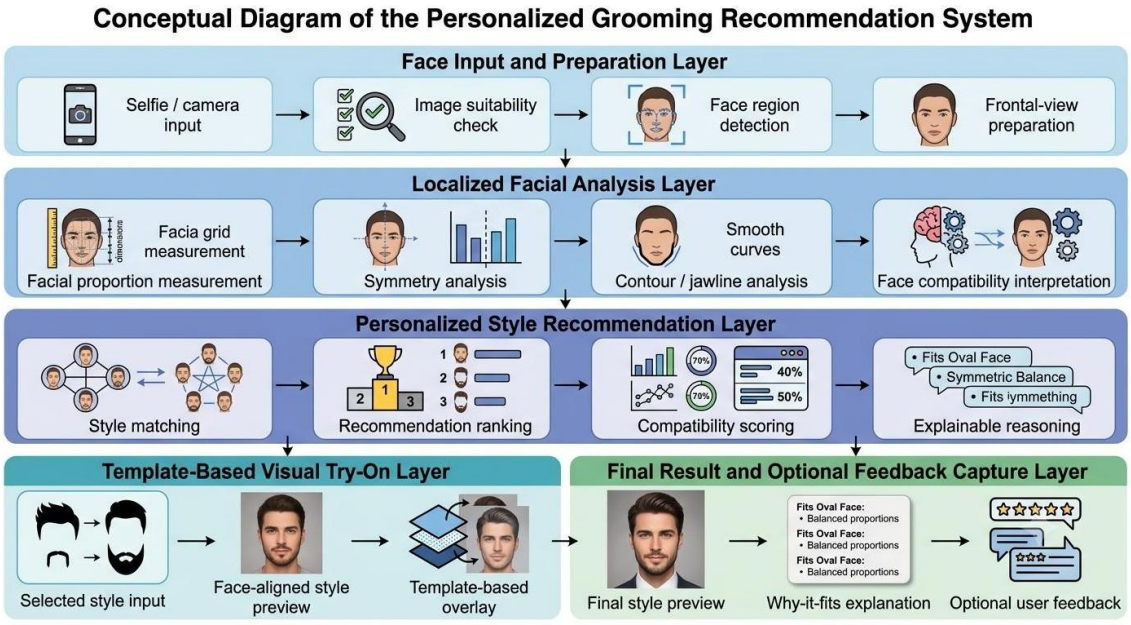
Process:

The core processing stage performs localized facial analysis by studying facial proportions, symmetry, contour, and jawline characteristics. These extracted facial attributes are then interpreted to understand face compatibility and style suitability. Based on this interpretation, the system generates personalized grooming recommendations by matching facial characteristics with appropriate style options and ranking them according to compatibility. The selected style is then presented through a template-based visual try-on stage, where the recommended grooming style is previewed in an understandable way. The system also generates explainable reasoning so users can understand why a particular style is recommended for them.

Output:

The final output consists of a style preview, a personalized recommendation result, and a short explanation of why the selected style fits the user's facial features. In addition, the system may capture optional user feedback. This output supports both user understanding and practical grooming decision-making.

As illustrated in the below image, the system is structured into five sequential layers: Face Input and Preparation, Localized Facial Analysis, Personalized Style Recommendation, Template-Based Visual Try-On, and Feedback Capture. This unified, production-ready pipeline directly translates raw image input into an actionable, tailored grooming consultation. By combining automated quality gatekeeping, localized anthropometry, and generative try-ons, the platform seamlessly bridges the divide between theoretical computer vision frameworks and actual retail software.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Acquisition of Specialized Domain Expertise

We can gain the specialized domain expertise for this project from a focused set of research papers, review articles, and practical grooming-related references. For South Asian facial structure and anthropometric grounding, we can take information from ***The South Asian facial anthropometric profile: A systematic review*** *(PubMed) and* ***Anthropometric Analysis of Indian Faces: A Systematic Review*** *(PubMed). For hairstyle and face-shape recommendation logic, we can take ideas from* ***A hybrid approach to building face shape classifier for hairstyle recommender system*** *(ScienceDirect) and* ***Automatic Face Shape Classification Via Facial Landmark Measurements*** *(R Discovery). For practical face-shape mapping and recommendation ideas, we can also take supporting concepts from* ***Hairstyle Recommendation Based on Face Shape Using Machine Learning*** *(KDU repository) and related facial-feature-based recommendation papers.*

Acquisition of Technical Knowledge and Skills

The technical workflow requires competencies spanning mobile application development, real-time computer vision, and generative machine learning. For facial landmark extraction and mesh generation, the team will adopt code structures and usage patterns from the Google MediaPipe Face Mesh Documentation to support high-density facial coordinate mapping. For facial measurement and shape interpretation, practical execution patterns will be adapted from open source OpenCV and NumPy scripts that programmatically calculate geometric distances, facial proportions, and structural symmetry indices.

For face-shape mapping logic and recommendation architecture, the system incorporates concepts from regional academic baselines, specifically modifying rules documented in Hairstyle Recommendation Based on Face Shape Using Image Processing from the KDU Institutional Repository. Finally, for the generative try-on phase, the framework utilizes practical, face-aligned preview pipelines built on open-source repositories of ControlNet and IP-Adapter, customizing these pre-trained diffusion workflows into an interactive, training-free personal grooming application.

Data Collection Strategy

For data collection, publicly available image datasets from ***Kaggle*** *are suitable as the primary source for initial development and testing. These datasets can provide facial images with variation in face shapes, facial structures, and pose, which supports the early stages of model validation and component testing. Since a fully Sri Lankan-specific dataset is limited, a South Asian-based public dataset can serve as the baseline for calibration and refinement, allowing the system to remain regionally relevant while still being practical within the project scope.*

This gives the project a clear and feasible data direction for experimentation, validation, and improvement.

Ethical Considerations and Clearance

This project does not require ethical clearance because it is a general, non-invasive grooming recommendation system and not a medical or diagnostic application. It does not involve high-risk decision-making, clinical evaluation, or harmful personal data use.

9. Objectives and Novelty

Main Objective *To design and develop an AI-assisted personalized grooming recommendation system that analyzes the facial characteristics of South Asian, particularly Sri Lankan, young adults, and provides explainable hairstyle and grooming recommendations through quality-aware facial profiling, localized face-shape interpretation, ranked style matching, and a template-based & descriptive visual preview.*			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Ranaweera H.H.E.U. IT23406308	*Component 1: Quality-Aware Facial Profiling and Landmark Extraction*	• *Check the input image quality by assessing clarity, lighting, blur, and overall suitability before any analysis begins.* • *Detect the face region from the image and validate whether the head pose is appropriate for further processing.* • *Extract facial landmarks and use them to identify key facial contour points, facial structure details, and hairline-related features.* • *Measure important facial contour and hairline characteristics so that only reliable facial information is passed to the next stage.*	*Image-quality-aware gatekeeping that prevents poor inputs from affecting later stages, along with contour and hairline profiling that gives the system stronger facial understanding than basic face detection.*

Kodithuwakku P.R.M. IT23491526	Component 2: *Facial Analysis, Scoring, and Interpretation*	• *Calculate facial proportions and analyze the geometric relationships between major facial features.* • *Classify the face shape based on the extracted facial measurements and structure.* • *Score facial symmetry and balance using the landmark-based profile created in the previous stage.* • *Combine the different measurements into a single grooming suitability score and generate a short explanation card for interpretation.*	*South Asian-focused facial interpretation with multi-metric scoring and explainable output, making the system more regionally relevant and transparent than generic face-shape tools.*
Nimesha A.W.P.T. IT23270992	Component 3: *Style Recommendation and Ranking Engine*	• *Utilize the structured facial profile from Component 2 to match user facial characteristics with predefined grooming style families and identify suitable hairstyle, beard, and facial grooming options* • *Apply hybrid rule-based filtering and weighted compatibility scoring using multiple facial attributes.* • *Rank styles using confidence-aware Top-N ranking based on face shape, facial proportions, jawline, forehead, hairline, and symmetry.* • *Provide separate recommendations for hairstyles, beard styles, and facial grooming styles.*	*Ranked Top-N grooming recommendations with confidence scores and explainable style matching using multi-attribute facial compatibility analysis and hybrid scoring-based ranking.*

Hewapathirana S.I. IT23286078	Component 4: *Style Preview and Result Presentation*	*• Perform face and hair/beard region segmentation and select the appropriate visual template based on Component 3's top-ranked style output.* *• Apply geometry-aware affine alignment to warp the selected template to the user's actual measured facial proportions from Component 1, rather than using a generic fixed overlay.* *• Convert Component 3's structured attribution data into a readable explanation card displayed alongside the preview (e.g., "Recommended because of your oval face shape, suits your jawline").* *• Render a before/after comparative preview showing the original face alongside the recommended style.* *• Capture user feedback (agree/disagree or rating) per previewed style as a structured signal to support future refinement of the ranking model.*	*Identity-preserving, geometry-aware visual try-on that aligns style templates to the user's own measured facial proportions rather than a generic overlay, combined with rendering of pre-generated explanation content into an easily understandable preview — making the system's recommendations visually verifiable and transparent rather than a simple static image swap.*

10. This section is to be completed by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Data required for the project is available.

Yes	✓	No	

f) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor*	Ms.	Tharushi	Rubasinghe	28/06/2026
Co-Supervisor*	Dr.	Dinuka	Wijendra	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

*- At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.